

Highlights

A Reproducible Benchmark and Cross-Domain Stress Test for Fragmentation-Event Forecasting in Unmanned Aerial Vehicle Networks

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- The locked primary test finds no supported source-gated superiority.
- A full factorial isolates current, graph, topology, and motion sources.
- Packet and controller experiments expose prediction-to-action limits.
- Cross-domain tests reveal severe fragmentation-event transfer failure.
- Runtime measurements bound claims to the host-side algorithm loop.

A Reproducible Benchmark and Cross-Domain Stress Test for Fragmentation-Event Forecasting in Unmanned Aerial Vehicle Networks

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Abstract

Rapid motion and correlated radio variation can fragment an unmanned aerial vehicle (UAV) network before reactive monitoring leaves time for a safe response. We test whether short-horizon warnings remain useful when false alerts, packet traffic, controller cost, domain shift, and computation are considered jointly. The artificial-intelligence contribution is a leakage-controlled benchmark of current-state, topological, kinematic, source-gated, shallow, and neural predictors with a locked 20-seed evaluation. The engineering application couples validation-selected warnings to bounded simulated relay and packet-event models; it is not field control. Under the prespecified multiplicity-controlled rule, Source-Gated Kinetic-TopoGuard does not outperform Current-state ExtraTrees: the paired event F1 difference is -0.0215 (seed-bootstrap 95% confidence interval [-0.0417, -0.0020]; Holm-adjusted permutation $p = 0.1030$), while false alerts increase by 2.10 events per minute. In the 20-seed equal-learner factorial, adding graph, topology, and kinematic sources reduces event F1 by 0.0426, although calibration improves. The residual count branch selects zero scale in 95% of seeds, and none of 360 paired closed-loop comparisons satisfies the predeclared engineering-benefit rule. The single compatible measured three-vehicle ultra-wideband sequence contains only one or two fragmentation events per threshold; all frozen and adapted models yield zero event F1 at the primary threshold. We treat this as a sparse stress test, not strong external validation. Measured host-loop latency excludes sensing, telemetry, autopilot communication, and actuation. The benchmark exposes where richer predictors fail to yield engineering benefit.

Keywords: Fragmentation-event forecasting, Unmanned aerial vehicle networks, Reproducible benchmarking, Cross-domain stress testing, Network

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1. Introduction

Flying ad hoc networks (FANETs) support cooperative sensing, mapping, surveillance, and emergency response without requiring fixed communication infrastructure. Their topology changes more rapidly than that of terrestrial mobile networks because aircraft move in three dimensions, execute mission turns, and can separate at tens of metres per second [1, 2]. A controller that learns about a partition only after links disappear is therefore reactive. A useful predictor must estimate whether the currently connected swarm will fragment at a future control horizon, issue a warning before the transition, and expose enough structure for an engineer to assess false-alert cost.

The zeroth Betti number, β_0 , is the number of connected components. It gives a direct network interpretation: $\beta_0 = 1$ denotes a connected graph and $\beta_0 > 1$ denotes fragmentation. Exact graph traversal or disjoint-set algorithms compute β_0 for an observed graph [3, 4], but they do not forecast its future value. Persistent homology complements a single-radius graph by recording how components merge as the distance threshold grows. A persistence image turns those merge scales into a stable fixed-dimensional feature [5]. In a mobile network, however, velocity determines whether a marginal link is closing or opening. The engineering question is not merely whether these descriptors can be computed, but whether they add reliable decision value beyond a strong current-state comparator.

Graph neural networks (GNNs) have been applied to routing, clustering, and coverage [6, 7, 8], while topology-aware networks combine message passing with persistent-homology descriptors [9, 10]. These studies do not establish event-level early warning under matched learners, unused-seed confirmation, packet-level control, and cross-domain transfer. Randomly mixed samples can also leak neighbouring states from one flight into training and testing. The gap is therefore an evidence problem: model families, feature sources, operating thresholds, and engineering outcomes must be compared without letting held-out tests select the story.

The study addresses three research questions:

1. Under run-grouped and locked-seed evaluation, do graph, topological, kinematic, or source-gated predictors improve fragmentation-event F_1 over Current-state ExtraTrees?
2. Do improvements in forecast or calibration metrics survive chronological public-data transfer and produce paired packet/controller benefit at acceptable action cost?
3. Which computational and evidence boundaries prevent host-side simulation results from being interpreted as onboard, field, or deployment-ready performance?

The contributions follow the benchmark/stress-test route selected from the completed evidence:

1. **Evaluation formulation:** future fragmentation risk is separated from component-count regression, and one-to-one event matching prevents prolonged fragmentation or duplicate warnings from inflating performance.
2. **Artificial-intelligence benchmark:** current-state, graph, persistent-homology, kinematic, source-gated, shallow, and neural model families are compared with run-grouped splits, a 20-seed equal-learner factorial, and a locked 20-seed test unused in development.
3. **Engineering application:** validation-selected warnings drive a causal, physically bounded relay policy whose graph outcomes are kept separate from packet-event delivery, delay, drop reasons, and action cost.
4. **Cross-domain stress test:** field motion, UAV-to-infrastructure cellular measurements, held-out peer-link measurements, and three-UAV ultra-wideband topology are assigned explicit and non-overlapping claim boundaries.
5. **Reproducible negative evidence:** source-induced event degradation, an inactive residual branch, weak measured-topology transfer, absent closed-loop support, and model-dependent host latency are retained rather than selected away.

2. Related work

Future fragmentation and link forecasting. GCN, GAT, and GraphSAGE aggregate neighbouring node state [11, 12, 13]; FANET and non-terrestrial-network studies mainly apply such encoders to routing, clustering, or policy learning [6, 7, 8, 14, 15]. Recent UAV link prediction combines multiscale attention with recurrent temporal state [16]. Those tasks predict links or actions, whereas this study predicts the first future transition from one connected component to multiple components. That target requires a system-level comparator: a link predictor can improve edge accuracy without improving the timing or false-alert burden of a fragmentation warning.

Temporal and topology-aware learning. Recurrent graph convolution, STGCN, and continuous-time TGN provide distinct temporal inductive biases [17, 18, 19]. PH-GNN, PersLay, and persistence images expose global multiscale structure that shallow message passing can miss [9, 10, 5]. For H_0 , finite merge times are minimum-spanning-tree edge lengths, making the descriptor tractable for UAV swarms. The unresolved task-specific question is whether these sources add held-out event value after recent component state and kinematics are controlled with the same learner. The equal-learner factorial and the contextual neural extension are designed around that comparison rather than an architecture-only leaderboard.

Leakage-safe event evaluation. Neighbouring snapshots from one flight are strongly dependent, so random row splits can expose a model to the same trajectory in training and test. The present protocol groups complete runs and

freezes validation-selected thresholds. It also adapts range-aware time-series detection principles [20] to a one-to-one, pre-transition matching rule. This distinction matters because sample F_1 can reward prolonged fragmented periods, while the engineering endpoint concerns one early warning per transition and the number of unmatched warnings per minute.

From prediction to engineering evidence. Public sources observe different links in the causal chain: forestry flights provide motion without peer-radio labels [21]; AERPAW provides aerial cellular KPI and throughput [22, 23]; WiNES provides measured UAV-to-UAV channel samples [24, 25]; and MILUV combines indoor motion capture with three-robot UWB measurements [26, 27]. None alone validates outdoor FANET warning-to-packet-to-actuation performance. Accordingly, this work separates forecast metrics, measured-data transfer, packet-event simulation, action cost, and host-loop latency instead of treating any one layer as deployment evidence.

3. Method

3.1. Physical graph and forecast target

At time t , UAV positions and velocities are $P_t = \{p_i(t)\}_{i=1}^N$ and $V_t = \{v_i(t)\}_{i=1}^N$. A geometric policy first supplies fixed and adaptive candidate radii. Candidate links are then filtered by a bidirectional received-power rule:

$$(i, j) \in E_t \iff d_{ij}(t) \leq r_t, \quad P_{i \rightarrow j}(t) \geq \Gamma, \quad P_{j \rightarrow i}(t) \geq \Gamma. \quad (1)$$

The link budget uses log-distance path loss, log-normal shadowing, Rayleigh or Nakagami fading, and transmitter/receiver hardware offsets, following standard radio abstractions and the established correlated-shadowing and Nakagami models [28, 29, 30]. Hardware offsets are fixed within a run. Shadowing and fading evolve as Gauss–Markov processes, so consecutive radio snapshots are temporally correlated rather than independent draws. The same realization is used for paired fixed/adaptive-policy comparisons.

Table 1 makes the physical and control assumptions auditable. Literature-backed model forms are distinguished from study-specific stress settings; none of the numerical settings is presented as a calibrated radio, certified airframe, or regulatory limit.

For a horizon of h ticks, the count target and sample-level risk label are

$$y_t = \beta_0(G_{t+h}), \quad z_t = \mathbb{K}[\beta_0(G_{t+h}) > 1]. \quad (2)$$

The event target is stricter. An event exists only when

$$\beta_0(G_{t-1}) = 1 \quad \wedge \quad \beta_0(G_t) > 1. \quad (3)$$

Thus repeated fragmented samples do not create repeated ground-truth events.

Table 1: Radio, forecast, and controller parameters. “Stress setting” denotes a fixed simulation choice rather than an empirical hardware claim.

Domain	Parameter	Value	Status and source
Radio propagation	Path-loss exponent; shadowing SD	2.35, 3.0 dB (nominal); 2.65, 4.5 dB (degraded)	Scenario values in the log-distance and correlated-shadowing abstractions [28, 29]; not site-calibrated measurements.
Small-scale fading	Nakagami m ; temporal correlations	$m = 1$; $\rho_{\text{shadow}} = 0.96$; $\rho_{\text{fade}} = 0.80$	$m = 1$ gives Rayleigh-equivalent severity in the Nakagami model [30]; correlations are stress-test settings.
Link budget	P_{tx} ; antenna gains; system loss; margin; sensitivity	20 dBm; 5 + 5 dB; 2 dB; 10 dB; −96/−92 dBm	Fixed simulation protocol; values are not hardware certification or spectrum-authorization claims.
Interference abstraction	Transmission probability; noise floor; SINR threshold	0.35; −101 dBm; 8 dB	Packet/radio stress settings used identically in paired comparisons.
Forecast and alerting	Tick; horizon; default refractory period	0.1 s; 6 ticks (0.6 s); 6 ticks	Prespecified evaluation protocol; window and refractory sensitivity is reported separately.
Relay controller	Maximum speed; acceleration; link-budget boost	30 m.s ^{−1} ; 12 m.s ^{−2} ; 3 dB	Bounded simulated intervention, motivated by aerial-relay positioning [31]; not airframe limits.
Relay state machine	Reaction; action; cooldown	2, 12, 6 ticks	Fixed controller stress settings (0.2 s, 1.2 s, 0.6 s).

3.2. Motion-conditioned topology features

Constant-velocity projection gives

$$\hat{p}_i(t + h) = p_i(t) + h\Delta t v_i(t). \quad (4)$$

From current and projected positions, the method extracts pair-distance quantiles, minimum margins to the active radius, closing/opening rates, projected component counts, and the number of links expected to cross the radius. Current graph features contain degree, density, clustering, path, and component summaries.

For H_0 persistence, pairwise distances define a Vietoris–Rips filtration. Finite component deaths are the minimum-spanning-tree merge lengths. Gaussian kernels on birth and persistence coordinates form a persistence image π_t . The implementation uses only H_0 ; it does not claim loop or cavity information.

The complete feature vector is

$$x_t = [c_t, g_t, \pi_t, k_t], \quad (5)$$

where c_t is current/lagged component context, g_t is current graph structure, π_t is the persistence image, and k_t contains kinematic projections.

3.3. Comparators, residual prediction, and risk scoring

Two current-state references have different roles. *Current-state persistence* is deterministic: it sets $\hat{y}_t = \beta_0(G_t)$ and does not fit a learner. It must not be confused with *Current-state ExtraTrees*, which fits an ExtraTrees residual regressor and a balanced ExtraTrees risk classifier from current and lagged component state, component change, node count, and active radius. The latter uses the same run split, validation threshold rule, and test evaluation as the richer equal-learner factorial rows. The fitted implementation uses the original Extremely Randomized Trees method through scikit-learn [32, 33].

Kinetic-TopoGuard anchors count prediction at the observed component count:

$$\hat{y}_t = \max\{1, \beta_0(G_t) + \alpha f_\theta(x_t)\}. \quad (6)$$

An ExtraTrees regressor estimates the residual. The scale α is selected from a grid that includes zero, using validation mean absolute error (MAE) only; the test partition cannot influence whether the residual branch is active. A separate balanced ExtraTrees classifier gives $s_t = P(z_t = 1 \mid x_t)$. Classifier choice, score blending, and the alert threshold are selected on validation runs only.

3.4. Source-Gated Kinetic-TopoGuard

Source-Gated Kinetic-TopoGuard tests whether learned source selection is safer than forced feature concatenation. Four base sources are used: current state, graph summaries, zero-dimensional persistence features, and kinematic projections. Within the training partition, GroupKFold cross-fitting holds out complete split groups. Every training row therefore receives count-residual and fragmentation-risk predictions from base learners that were not fitted on that row, and no run identifier appears on both sides of one fold. The four out-of-fold count predictions train a ridge meta-regressor; the four out-of-fold risk scores train a regularized logistic meta-classifier. Base learners are then refitted on the full training partition for inference.

Development validation runs select minimum leaf sizes, ridge regularization, logistic regularization, probability calibration from none, sigmoid, or isotonic regression, and the event threshold. Calibration is evaluated with reliability diagnostics, Brier score, and expected calibration error [34, 35]. Threshold selection maximizes validation event F_1 , then sample F_1 , then minimizes false events per minute, and finally prefers the higher threshold. The test partition is evaluated once after these choices. Stored artefacts contain base parameters, meta coefficients, calibration type, threshold, feature-source names, and cross-fitting run identifiers. For the locked confirmation, the modal validation-selected development package is frozen before any new-seed test result is observed.

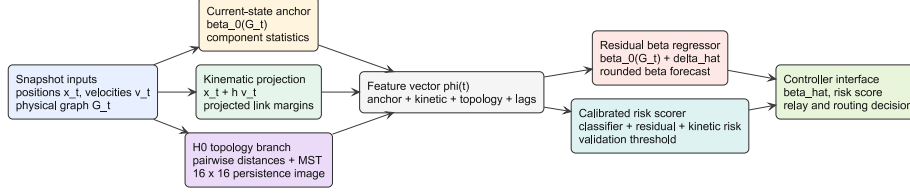


Figure 1: Kinetic-TopoGuard information flow. The controller interface consumes the forecast and risk score. Both the analytical one-step test and the stateful packet-coupled policy are physically constrained simulations rather than live command paths.

For each ground-truth transition at index t , eligible alerts are those emitted while the graph is connected in $[t - h, t - 1]$. Events and alerts are processed chronologically; one eligible alert can match at most one event, one event can receive at most one match, and alerts emitted after fragmentation begins are ineligible. Unmatched eligible alerts are false-alert events. The study reports event precision, recall, F_1 , missed events, and false-alert events per simulated minute. This explicit one-to-one rule is study-specific; range-aware time-series precision and recall provide the broader evaluation context [20]. Sample-level F_1 , precision-recall area, receiver-operating-characteristic area, Brier score, and expected calibration error remain complementary diagnostics rather than substitutes for the event endpoint.

Inference sequence.. Given P_t, V_t, G_t , horizon h , fitted f_θ, q_ϕ , and threshold τ : (1) project positions to $\hat{P}_{t+h}$; (2) compute current-graph, kinematic, and H_0 persistence features; (3) calculate $\hat{y}_t = \max(1, \beta_0(G_t) + \alpha f_\theta(x_t))$ and $s_t = q_\phi(x_t)$; and (4) emit an alert only if $\beta_0(G_t) = 1$ and $s_t \geq \tau$.

3.5. Controller and packet-simulation boundary

The analytical one-step controller tests graph reachability after a bounded relay change; its output is a *reachability-delivery proxy*, not packet delivery ratio. The separate stateful closed-loop experiment carries relay position, velocity, reaction delay, action duration, and cooldown across controller ticks. Relay motion has maximum speed 30 m.s^{-1} and maximum acceleration 12 m.s^{-2} , and only relay-incident links receive at most a 3 dB budget improvement. This bounded repositioning study is related to aerial-relay connectivity maintenance [31]. The coupled SimPy discrete-event model [36] separately generates packet arrivals, queues, routes, serialization, deadlines, retransmission outcomes, and explicit drop causes. Only delivery events from that packet simulator are called packet delivery ratio. Neither controller is a flight-control implementation.

4. Experimental protocol

4.1. Simulation and leakage control

Mission trajectories combine scan, transit, and loiter phases. Swarm sizes, mobility families, fixed/adaptive graph policies, and nominal/degraded radio

scenarios are crossed within each seed. Complete base runs, identified by mobility, swarm size, and seed, are assigned to train, validation, or test. Policy and radio variants of one trajectory remain in the same split. No test sample selects a learner, residual scale, risk threshold, or alert budget. Training, validation, and test run-identifier intersections are checked as empty for every seed. All paired models within a seed share the same mobility and radio realization.

The original 20-seed benchmark compares Kinetic-TopoGuard, validation-selected shallow machine learning, and deterministic current-state persistence. The equal-learner factorial adds Current-state ExtraTrees and all eight combinations of graph, topology, and kinematic sources. The five-seed model-family extension contains the same five non-neural references and nine PyTorch baselines: graph convolutional, graph attention, GraphSAGE, persistence-image multilayer perceptron, two FANET-TopoGNN variants, temporal graph convolutional, spatio-temporal graph convolutional, and temporal graph network models. They use PyTorch [37]; the repository also exposes an optional PyTorch Geometric adapter [38], although PyTorch Geometric was not installed in the measured five-seed runtime. Every neural aggregate requires the same five seeds and a verified PyTorch backend; surrogate or incomplete rows are rejected. Because these neural rows have only five seeds, they are descriptive model-family context: they are not included in the locked hypothesis, do not support superiority claims, and are not pooled with any 20-seed aggregate. The full Supplementary Material S1 table reports all 14 models, parameter counts where defined, training budgets, event metrics, and seed coverage. A separately documented 20-new-seed extension is pending execution on a verified PyTorch backend; this manuscript reports no result for that unexecuted protocol.

After the five-seed Source-Gated development study, a separate confirmation is locked by SHA-256 hash. Its 20 seeds are generated by a domain-separated SHA-256 counter stream and are rejected if they appear in any prior configuration or output. Before those test outcomes are read, the feature definitions, horizon, base and meta hyperparameters, isotonic calibration, threshold grid and tie-break, primary comparator, primary event- F_1 hypothesis, and evaluation models are frozen. The lock includes the config hash, source artefact hashes, excluded prior seeds, and git commit. A hash mismatch aborts execution; each test partition is evaluated once.

MAE and R^2 confidence intervals use clustered bootstrap resampling of complete runs. Risk metrics are computed from held-out samples. Event metrics use the transition matching defined in Section 3. The two operating policies impose validation budgets of two and one false alert events per minute. Feasibility is recorded rather than assumed. For the locked primary event- F_1 comparison, the seed-bootstrap interval is reported as an effect-size uncertainty summary, while the paired permutation test supplies the confirmatory p -value. The prespecified claim requires a positive paired difference, a bootstrap interval excluding zero, and Holm-adjusted permutation $p < 0.05$ across the locked primary comparisons. When these summaries differ, the multiplicity-controlled rule governs the superiority decision; no single unadjusted or interval result is used to override it.

4.2. Stress, ablation, packet, and latency studies

The horizon study reuses each physical trajectory and radio realization and deterministically reindexes labels at 0.2, 0.4, 0.6, 1.0, 1.5, and 2.0 s. Five seeds are used at every horizon. The feature study crosses graph, topology, and kinematic sources in a complete 2^3 factorial. Every row uses the same ExtraTrees learner and current-state context, preventing a stronger classifier from being attributed to one feature family.

The event-definition sensitivity keeps the fitted Kinetic-TopoGuard score and its validation-selected threshold fixed. On held-out runs it varies the pre-transition warning window and refractory period over 0.3, 0.6, and 1.2 s, and reports results across within-seed low, middle, and high measured event-density tertiles. These analyses test the stability of event accounting; they do not retune the model or replace the locked endpoint.

The packet-load sweep evaluates 8, 16, 32, and 64 offered packets per 0.1 s tick. Models receive identical source–destination demand and random seeds. It implements finite queues, route availability, CSMA-style random backoff, a single collision domain, and a per-tick deadline. This is a controlled packet abstraction, not ns-3, OMNeT++, or a field MAC trace.

The stateful closed-loop study instead crosses three packet loads and swarm sizes of 10, 20, and 30 for six policies. Mobility, radio, packet arrivals, and packet random seeds are paired. A policy reads only current/past state and its current risk score; thresholds and action parameters are fixed from validation or a priori limits. Engineering benefit requires a packet-delivery-ratio confidence interval excluding zero, Holm-adjusted paired $p < 0.05$, relay travel below 300 m per simulated minute, and action duty cycle below 0.25.

Host-loop timing separates pairwise distance, radio graph, graph statistics, persistence image, feature assembly, count inference, risk inference, calibration, and controller decision. After five warm-ups, 30 iterations are measured at 10, 20, 30, and 40 UAVs under single- and default-thread modes. Sensor acquisition, telemetry, scheduling outside the process, autopilot communication, actuation, and network transfer are excluded.

4.3. External evidence layers

Forestry motion. Three synchronized public flight trajectories are sampled at 10 Hz. Because the bags contain no peer RF or packet labels, communication radii are predeclared counterfactual sensitivities. Models and thresholds are frozen from simulation.

AERPAW cellular. LTE connection state is predicted with a chronological final-30% test split and compared with a validation-derived RSRP threshold. iPerf throughput uses both final-30%-by-time and first-half-to-second-half splits. The lag-1 throughput autocorrelation is reported, and interleaved random blocks are prohibited.

60 GHz UAV-to-UAV channel. A measured peer-link model is trained below 33 m and tested on held-out distances at or above 33 m. Link viability uses the same SNR and received-power rule in model testing and downstream

sensitivity. Applying the measured distance-success relationship to the forestry motion is a *transported RF sensitivity*, not same-site calibration. Primary summaries require all three pair distances to remain inside the measured 6–40 m support; clamped full-trace values are secondary.

MILUV UWB topology. The official Figshare archive for `cirObstacles_3_random_0` is accessed by HTTP range extraction. Only six Vicon and UWB CSV members are retained, with archive CRC and SHA-256 verification. Motion is interpolated at 10 Hz; reciprocal ranging records are deduplicated; and each robot-pair edge uses the latest causal first-path-power observation no older than 6 s. The longest interval with all three pair states known is evaluated at -90 , -92 , and -95 dBm. No MILUV sample is used for fitting or threshold selection; the three-threshold sensitivity grid is fixed before model evaluation and is not optimized against MILUV outcomes. These labels describe measured UWB ranging quality, not IP PDR. The evaluated sequence contains two connected-to-fragmented events at -90 and -92 dBm and one at -95 dBm. This event support is sufficient only for a sparse failure-oriented stress test, not for a precise external-validation estimate.

The frozen zero-shot MILUV result remains the primary transfer evidence. A secondary chronological study uses an early calibration prefix, a guard interval, and a later test segment. Base weights remain frozen for probability calibration and threshold selection; a separately labelled few-shot protocol may update only its predeclared prediction head from the prefix. Adjacent samples are never randomly mixed. Leave-one-scenario-out testing is reported unavailable because only one compatible provenance-verified three-robot sequence is present.

5. Results

5.1. Locked primary confirmation and benchmark context

The locked comparison is reported first because it is the only evaluation whose 20 seeds were excluded from all prior configurations and outputs. Table 3 shows the complete locked aggregate. Current-state ExtraTrees obtains event $F_1 = 0.498$, whereas Source-Gated Kinetic-TopoGuard obtains 0.477. The paired Source-Gated-minus-current difference is -0.0215 (seed-bootstrap 95% CI $[-0.0417, -0.0020]$); the paired permutation p -value is 0.0515 and its Holm-adjusted value is 0.1030. The predeclared superiority rule is therefore not met. The interval summarizes the observed paired effect, whereas the Holm-adjusted permutation result controls the confirmatory decision; the apparently different thresholds are not treated as interchangeable evidence. Event precision decreases by 0.0831 and false alerts increase by 2.10 events/minute (95% CI $[1.12, 3.07]$, Holm-adjusted $p = 0.00249$). Although the Source-Gated model improves Brier score and expected calibration error, those secondary gains do not overturn the negative primary event decision.

The earlier 20-seed benchmark remains useful as protocol context rather than the primary superiority test; its complete table is in Supplementary Material S1. Event F_1 penalizes duplicate and unmatched alerts and exposes missed

Table 2: External evidence matrix. Direct measurements are separated from derived or transported quantities and their permitted interpretation.

Evidence	Direct measurement	Derived/transported quantity	Allowed claim	Boundary
Forestry multi-drone field experiment	GNSS latitude; longitude; altitude; timestamps	aligned local ENU trajectories and pair distances	measured three-UAV field motion	measured peer RF; packet delivery; measured connectivity
Forestry multi-drone field experiment	motion only	radius graphs; component labels; frozen simulation-model predictions	motion-domain sensitivity under explicit radius assumptions	measured RF; measured PDR; field-validated connectivity
AERPAW Datasets 22/23 LTE	LTE connection state; RSRP; RSRQ; RSSI; position; altitude; timestamps	local geometry; RSRP threshold; logistic connection probability	chronological UAV-to-infrastructure LTE connection-state prediction	inter-UAV validation; FANET PDR
AERPAW Dataset 23 iPerf	iPerf throughput; LTE KPIs; position; timestamps	timestamp-aligned RF/KPI throughput predictions	chronological throughput prediction including negative R2	robust throughput transfer; inter-UAV PDR
WiNES UAV-to-UAV 60 GHz	UAV-to-UAV distance; altitude; TX/RX beam indices; SNR; received power; path loss; gain indices; link direction	link viability and held-out-distance model scores	held-out-distance UAV-to-UAV 60 GHz link viability	packet delivery; forestry same-site calibration
WiNES RF transported to forestry motion	WiNES RF at source site; forestry motion at separate site	interpolated pair-link probabilities and expected component count	in-support transported RF sensitivity	same-site calibration; measured forestry RF; packet delivery
WiNES RF transported to forestry motion	WiNES RF at source site; forestry motion at separate site	nearest-support clamping and expected component count	secondary clamped sensitivity only	extrapolative validation; same-site calibration; measured forestry RF

transitions. Deterministic persistence has the lowest count MAE (0.132 versus 0.133 for Kinetic-TopoGuard), while Kinetic-TopoGuard raises event F_1 from 0.414 for shallow machine learning to 0.483 and reduces false alerts from 13.24 to 3.53 per minute. This is a warning-trade-off result against the shallow comparator, not evidence that the residual count branch or richer sources outperform current-state models.

The five-seed neural extension is retained as model-family context rather than folded into the 20-seed claim. This distinction avoids implying that every expensive neural baseline was run for 20 seeds. Kinetic-TopoGuard obtains MAE 0.120 and event $F_1 = 0.470$. STGCN has the lowest neural MAE (0.218), while TGN has the highest neural event F_1 (0.464) but 6.39 false events/minute versus 3.30 for Kinetic-TopoGuard. Backend provenance and per-seed rows are included in the package. The compact table is selected by a rule fixed independently of whether a result is favourable: the primary non-neural methods are retained, together with the neural model having the lowest MAE and the neural model having the highest event F_1 (model name breaks ties). The supplementary full table reports all 14 executed baselines, their five-seed coverage,

Table 3: Locked 20-seed evaluation on previously unused seeds. Confidence intervals are seed-level intervals; thresholds are selected on validation runs only.

Model	Event F_1 [95% CI]	False/min	MAE	Brier
Current persistence	0.000 [0.000, 0.000]	0.00	0.139	0.077
Current-state ExtraTrees	0.498 [0.485, 0.511]	4.23	0.140	0.078
Kinetic-TopoGuard	0.473 [0.449, 0.497]	4.06	0.140	0.078
Source-Gated Kinetic-TopoGuard	0.477 [0.458, 0.495]	6.34	0.172	0.056
Shallow ML	0.397 [0.369, 0.426]	15.79	0.219	0.612

Table 4: Five-seed model-family extension under the predeclared compact-table selection rule.

Model	Seeds	MAE	Event F_1	False/min
Current-state persistence baseline	5	0.120	0.000	0.00
Current-state ExtraTrees	5	0.120	0.494	4.30
Kinetic-TopoGuard	5	0.120	0.470	3.30
Source-Gated Kinetic-TopoGuard	5	0.148	0.445	7.00
STGCN (w=5)	5	0.218	0.404	11.80
TGN (w=5)	5	0.234	0.464	6.39

event precision/recall/ F_1 , false-event rate, confidence intervals, backend, and model-budget metadata; failed or missing runs are reported separately rather than removed from aggregation.

5.2. Auditable feature contribution

We use *auditable post-hoc decomposition*, rather than claiming intrinsic model interpretability. Models and operating thresholds are frozen before the held-out test runs are explained. Grouped permutation moves complete equal-length run blocks, preserving within-run temporal dependence; TreeSHAP is an optional secondary view of the selected tree components [39, 40]. Neither explanation is used for feature, threshold, calibration, or model selection.

The detailed Supplementary Material S1 table reports the change in event F_1 after each group is permuted. Graph summaries are the highest-ranked group for both Kinetic-TopoGuard (mean drop 0.196, 95% CI [0.158, 0.244]) and Source-Gated Kinetic-TopoGuard (0.096 [0.055, 0.151]). Persistence-image bins contribute to Kinetic-TopoGuard (0.062 [0.012, 0.111]) but not detectably to the source-gated model (−0.0004 [−0.0217, 0.0178]). Projected-fragmentation statistics have smaller positive contributions in both models, whereas speed summaries and most pair-distance subgroups have intervals spanning zero. For component-count MAE, all Kinetic-TopoGuard grouped importances are zero in these five seeds; the frozen count prediction is therefore not presented as feature-driven. Source-gated count effects are dominated by current state, and pair-distance deltas show a small negative contribution. Negative and unsupported effects are retained in the machine-readable outputs and Fig. 2.

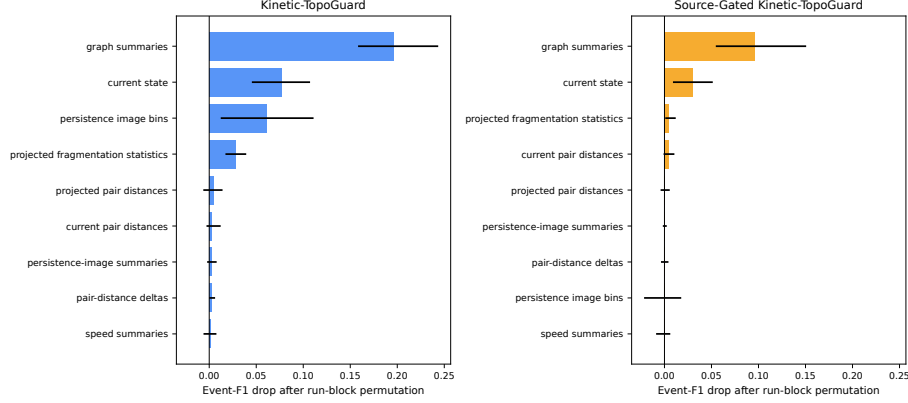


Figure 2: Grouped permutation importance with seed-bootstrap 95% intervals. Permutations exchange complete run blocks rather than individual, temporally dependent snapshots.

5.3. Horizon and feature-source sensitivity

The residual audit directly tests the count branch against its persistence anchor. The validation-selected scale is zero in 19 of 20 seeds (95%). Across held-out tests, selected minus persistence MAE is $+0.000650$ with a seed-bootstrap 95% CI $[0, 0.001949]$ and Wilcoxon $p = 0.317$. No seed improves, 19 are unchanged, and one worsens. Count forecasting is therefore retained as an audited component, not claimed as a primary contribution.

The Supplementary Material S1 horizon table prevents the 600 ms result from being interpreted as universal. Longer horizons change both class prevalence and the set of alerts that can match a transition, so event recall and event F_1 are reported alongside MAE and sample F_1 .

Table 5 shows that the event score is definition- dependent even with model and threshold fixed. Expanding the warning window from 0.3 to 1.2 s raises event F_1 from 0.368 to 0.552, whereas expanding the refractory period from 0.3 to 1.2 s lowers it from 0.555 to 0.363 while reducing false events. The middle-density stratum is especially weak ($F_1 = 0.151$); the low-density stratum has no ground-truth events and is reported as zero support, not evidence of successful detection. Thus the nominal 0.6 s endpoint is retained for comparability rather than selected from this descriptive sweep.

The 20-seed equal-learner factorial in Table 6 is deliberately stricter than a leave-one-feature-out ablation. Current-state ExtraTrees reaches event $F_1 = 0.514$; the full graph+topology+kinematic row reaches 0.471. The paired full-minus-current difference is -0.0426 (95% CI $[-0.0617, -0.0232]$, Holm-adjusted permutation $p = 0.0148$), and false alerts increase by 1.98 events/minute (95% CI $[1.17, 2.79]$, adjusted $p = 0.00874$). The richer row nevertheless improves Brier score by 0.0227 and expected calibration error by 0.0585, both with adjusted $p < 0.004$. The benchmark therefore exposes a calibration–event decision disagreement rather than topological or kinematic superiority.

Table 5: Five-seed held-out sensitivity of one-to-one event accounting. The fitted Kinetic-TopoGuard model and validation-selected threshold remain fixed. Intervals and per-seed values are supplied in Supplementary Material S1.

Sensitivity dimension	Level	Events/min	Precision	Recall	Event F_1	False events/min
Warning window	0.3 s	14.78	0.489	0.296	0.368	4.50
Warning window	0.6 s	14.78	0.625	0.377	0.470	3.30
Warning window	1.2 s	14.78	0.735	0.444	0.552	2.32
Alert refractory period	0.3 s	14.78	0.604	0.515	0.555	4.92
Alert refractory period	0.6 s	14.78	0.625	0.377	0.470	3.30
Alert refractory period	1.2 s	14.78	0.621	0.257	0.363	2.30
Measured event density	Low tertile	0.00	0.000	0.000	0.000	0.00
Measured event density	Middle tertile	3.15	0.166	0.144	0.151	2.20
Measured event density	High tertile	41.20	0.673	0.393	0.495	7.70

Table 6: Twenty-seed full 2^3 graph/topology/kinematic factorial; current-state context and the ExtraTrees learner are fixed in every row.

Feature sources	MAE	Sample F_1	Event F_1
Current state only	0.132	0.550	0.514
Graph	0.132	0.683	0.463
Graph + kinematic	0.132	0.665	0.463
Graph + topology	0.132	0.647	0.467
All sources	0.132	0.637	0.471
Kinematic	0.132	0.653	0.436
Topology	0.132	0.559	0.455
Topology + kinematic	0.132	0.513	0.441

5.4. Real-data transfer and scope

The forestry transfer tests whether a model trained on simulated mobility remains bounded on measured trajectories. It does not test measured radio. Raw and physically clipped predictions are both archived so that clipping cannot silently create the reported error. The counterfactual graphs contain five, two, and four connected-to-fragmented transitions at the three radii, respectively. All three models obtain zero event recall and event F_1 under their frozen thresholds. Low count MAE and high fragmented-sample F_1 must therefore not be read as successful early warning on this trace. Detailed per-radius results are retained in Supplementary Material S1.

The AERPAW LTE result is strong under chronological holdout: F_1 rises from 0.828 to 0.961 on Dataset 22 and from 0.679 to 0.980 on Dataset 23. Throughput is different. The random forest reduces MAE on the final-30% split but produces a strongly negative R^2 ; on the first-half-to-second-half split it is worse than the training-mean baseline. Together with lag-1 autocorrelation above 0.98, this is negative evidence against the earlier optimistic through-

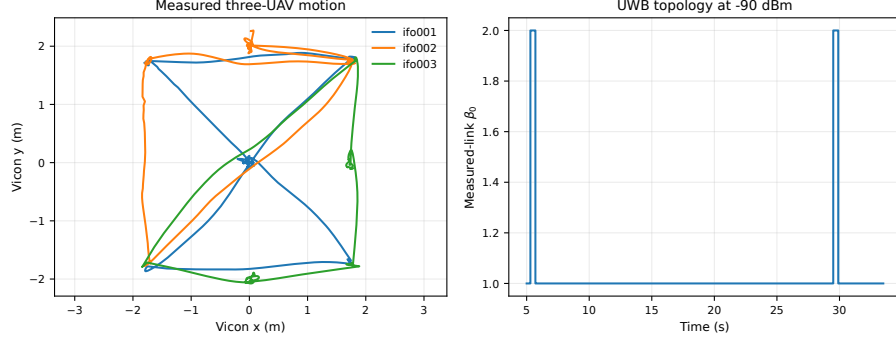


Figure 3: MILUV Vicon trajectories and the causally reconstructed measured-link component trace at the primary first-path-power threshold.

put claim. AERPAW therefore supports cellular link-state transfer, not robust throughput prediction or inter-UAV FANET validation. Split-specific metrics are supplied in Supplementary Material S1.

The measured 60 GHz logistic model improves held-out long-distance link F_1 from 0.731 to 0.923 and PR-AUC to 0.985. The forestry topology values in Supplementary Material S1 are reported only as transported sensitivities. Primary rows remain within measured distance support and use the same SNR-plus-received-power definition as the held-out link test.

MILUV provides simultaneous measured motion and inter-robot UWB topology for three flying vehicles, but only one compatible sequence and one or two connected-to-fragmented events per threshold. It is therefore used as a sparse cross-domain stress test rather than strong external validation. The original whole-sequence frozen sensitivity in Supplementary Material S1 is retained as the primary zero-shot artefact and is not mixed with adaptation. At the primary -90 dBm threshold, Kinetic-TopoGuard has count MAE 0.045 but event $F_1 = 0.08$; shallow machine learning reaches event $F_1 = 0.40$ while its count MAE exceeds 1.28. These are measured UWB topology-event outcomes, not outdoor range, packet delivery, or routing evidence.

The secondary chronological study, detailed in Supplementary Material S1, uses an early prefix, a guard interval, and an independent later test segment from the one compatible sequence. Zero-shot frozen, chronological calibration, and few-shot prediction-head rows remain separately labelled. At the same primary threshold, every model and all three protocols obtain event $F_1 = 0$. Calibration improves some Brier scores but provides no event- F_1 benefit; this is a negative transfer result, not an equivalence conclusion. A leave-one-scenario-out result is unavailable rather than synthesized from a single sequence. The repeated five-seed fits quantify algorithmic variability on this fixed trace; they do not increase the number of independent measured scenarios or fragmentation events.

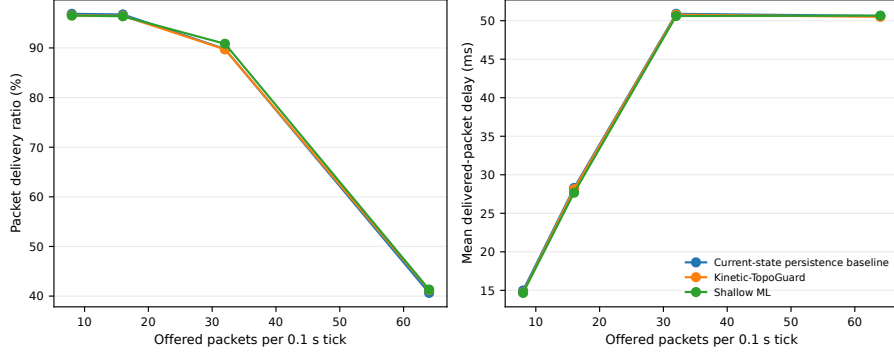


Figure 4: Packet delivery and delivered-packet delay versus offered load.

5.5. Operating points, packet behaviour, and latency

Operating points are selected per seed on validation event F_1 subject to one or two false event alerts per minute, then applied once to test runs. If a budget has no feasible threshold, the exported constraint flag records that fact; the analysis does not replace the policy with a test-selected threshold. Both budgets are feasible in all 20 validation seeds. On test runs, the two-event policy attains event $F_1 = 0.421$ at 1.85 false events/minute; the one-event policy trades recall for precision, yielding event $F_1 = 0.351$ at 0.90 false events/minute. The complete operating-point table is in Supplementary Material S1.

The one-step packet-load sweep, detailed in Supplementary Material S1, separates graph reachability from packet performance. At high offered load, contention-deadline drops dominate and all models degrade. Kinetic-TopoGuard does not consistently outperform the shallow or persistence controller. The correct conclusion is that better warning scores do not automatically translate into better PDR under a constrained MAC and one-step control policy.

The complementary stateful closed-loop experiment carries relay position, reaction delay, action duration, and action cost across controller ticks. Six policies share identical mobility, radio, source-destination packet arrivals, and packet random seeds. Unlike the reachability-delivery proxy, Table 7 reports packet events from the SimPy queue, routing, contention, and deadline model. The drop accounting is exhaustive: delivery, no-route, queue-overflow, deadline, link-failure, and intervention-transition outcomes sum to generated packets. Link and transition losses are retained as separate columns but are zero in the primary protocol because no measured within-tick or handover failure probability is available. Across the paired five-seed comparisons, neither original Kinetic-TopoGuard nor Source-Gated Kinetic-TopoGuard satisfies the joint confidence-interval, multiplicity-adjusted paired-test, and action-cost rule for consistent PDR superiority.

Table 8 times mutually exclusive stages of the host-side topology-to-warning

Table 7: Stateful closed-loop packet-event results across topology densities. Deadline success is conditional on a route existing at packet generation; latency quantiles use all delivered packets. Drop-rate columns use all generated packets.

Policy	Load	n	PDR	Deadline	P95 (ms)	Queue	No route	Queue drop	Deadline drop	Link/transition
Current-state ExtraTrees	8	5	0.963	1.000	27.55	3.63	0.037	0.000	0.000	0.000
Current-state ExtraTrees	16	5	0.963	1.000	53.00	8.21	0.037	0.000	0.000	0.000
Current-state ExtraTrees	32	5	0.909	0.943	91.41	17.69	0.036	0.000	0.055	0.000
No intervention	8	5	0.946	1.000	27.67	3.57	0.054	0.000	0.000	0.000
No intervention	16	5	0.946	1.000	53.22	8.10	0.054	0.000	0.000	0.000
No intervention	32	5	0.890	0.940	91.09	17.48	0.054	0.000	0.057	0.000
Kinetic-TopoGuard	8	5	0.964	1.000	27.72	3.63	0.036	0.000	0.000	0.000
Kinetic-TopoGuard	16	5	0.962	1.000	53.01	8.21	0.038	0.000	0.000	0.000
Kinetic-TopoGuard	32	5	0.907	0.942	91.52	17.69	0.037	0.000	0.056	0.000
Persistence-triggered	8	5	0.964	1.000	27.87	3.63	0.036	0.000	0.000	0.000
Persistence-triggered	16	5	0.963	1.000	53.43	8.21	0.037	0.000	0.000	0.000
Persistence-triggered	32	5	0.906	0.940	91.52	17.70	0.037	0.000	0.058	0.000
Reactive	8	5	0.961	1.000	27.89	3.62	0.039	0.000	0.000	0.000
Reactive	16	5	0.961	1.000	53.59	8.20	0.039	0.000	0.000	0.000
Reactive	32	5	0.901	0.937	91.52	17.70	0.039	0.000	0.060	0.000
Source-Gated Kinetic-TopoGuard	8	5	0.964	1.000	27.57	3.63	0.036	0.000	0.000	0.000
Source-Gated Kinetic-TopoGuard	16	5	0.963	1.000	52.79	8.21	0.037	0.000	0.000	0.000
Source-Gated Kinetic-TopoGuard	32	5	0.908	0.943	91.41	17.68	0.037	0.000	0.055	0.000

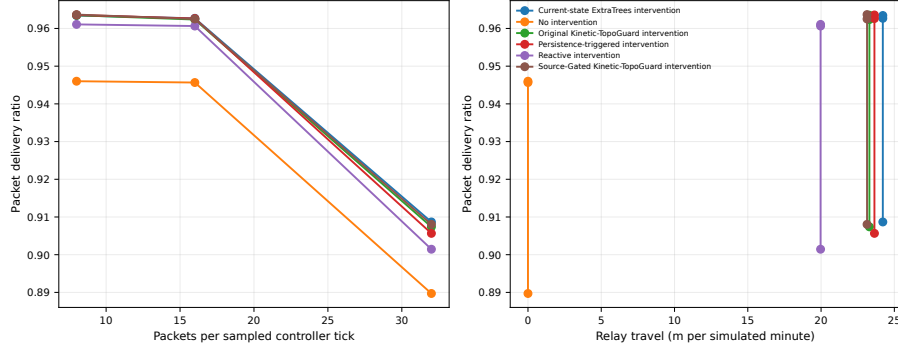


Figure 5: Closed-loop packet delivery versus offered batch load and relay travel. Values are packet-event outcomes, not graph-reachability proxies.

path and reports their additive total. Tail latency, not only model inference, is the relevant engineering quantity. At 30 UAVs with default library threading, P95 is 7.82 ms for FANET-TopoGNN and 13.23 ms for Kinetic-TopoGuard, but 44.52 ms for Current-state ExtraTrees and 90.81 ms for the Source-Gated variant. Thus, this measurement supports neither a model-independent 20 ms bound nor an onboard or sensor-to-actuator real-time claim. Sensor transport, operating-system scheduling outside the process, autopilot communication, and actuation remain outside the measured boundary. No Jetson hardware result is reported.

6. Engineering implications and deployment boundaries

The supported engineering decision is narrow: given an observed graph history, a frozen model can rank short-horizon fragmentation risk and issue a warning at an operating point selected on validation runs. The benchmark can then

Table 8: Measured additive host-loop timing at 30 UAVs (30 timed iterations after five warm-ups).

Model	Threads	P50	P95	P99 (ms)	Size (MiB)
Current-state ExtraTrees	default	43.31	44.52	44.64	0.18
Current-state ExtraTrees	single	43.14	44.18	44.90	0.18
FANET-TopoGNN	default	7.42	7.82	7.97	0.11
FANET-TopoGNN	single	8.17	8.75	8.91	0.11
Kinetic-TopoGuard	default	12.65	13.23	13.39	0.12
Kinetic-TopoGuard	single	12.69	13.16	13.56	0.12
Source-Gated Kinetic-TopoGuard	default	87.77	90.81	91.24	25.77
Source-Gated Kinetic-TopoGuard	single	89.66	92.60	93.26	25.77

compare whether a simulated, bounded relay response is preferable to no intervention or to a reactive rule. It does not support autonomous flight control, route certification, field packet-delivery guarantees, transmit- power authorization, or safe separation decisions. In particular, none of the 360 paired closed-loop comparisons meets the predeclared joint packet-delivery and action-cost rule.

Computation is similarly model-specific. On the measured host at 30 UAVs, default-thread P95 topology-to-decision latency ranges from 7.82 ms for FANET-TopoGNN to 90.81 ms for Source-Gated Kinetic-TopoGuard. The serialized Source-Gated artefact is 27,021,053 bytes, compared with 191,061 bytes for Current-state ExtraTrees and 121,272 bytes for Kinetic-TopoGuard. These local, trusted-artifact measurements establish neither a Jetson benchmark nor a sensor-to-actuator deadline. Edge deployment would additionally require profiling the chosen airframe computer, radio and telemetry ingestion, scheduling, power draw, failsafe communication, and actuator response.

Field validation requires synchronized multi-UAV motion, inter-UAV radio, medium-access, routing, packet, onboard timing, energy, and actuation observations under a preregistered intervention protocol. The included interactive interface is only a replay of generated and transported evidence; it is not part of the engineering-validation claim.

7. Discussion

The central result is not that more sources are always harmful, but that representation richness and engineering utility are different objectives. Current-state ExtraTrees is strong because the short forecast horizon, component history, component change, node count, and active radius already encode much of the immediately relevant state. Adding correlated or noisy summaries increases the ways in which validation-selected probability quality can diverge from one-to-one event precision and false-alert burden. The locked and factorial results show that better calibration alone is insufficient evidence of a better warning.

Feature contributions are conditional rather than universal. Run-block permutation assigns a positive event contribution to graph summaries in both au-

dated models, while persistence- image and speed intervals for the Source-Gated model include zero. Topological and motion descriptors may be useful when multi-scale structure or projected link closure is not recoverable from recent state, but the present seeds do not support that generalization. The residual audit gives the same warning for count regression: validation usually selects the physical persistence anchor, so the branch is an audited fallback rather than a central forecasting advance.

The measured-data layers also explain why simulation ranking cannot be transferred by assertion. Cellular connection state, peer-link viability, field motion, and indoor UWB topology observe different parts of the causal chain. Chronological MILUV calibration improves probability errors without recovering event detection, which motivates explicit domain calibration while showing that calibration cannot manufacture missing event signal. Likewise, closed-loop packet outcomes depend on traffic, queues, deadlines, action delay, and movement cost, not only warning F_1 . Retaining these disagreements and negative results turns the package into a falsifiable benchmark for future methods rather than a selected demonstration of one model.

8. Limitations

No available dataset jointly provides synchronized outdoor multi-UAV motion, inter-UAV radio frequency measurements, medium-access and routing state, packet delivery, onboard timing, and physical actuation. The primary training domain is therefore simulated, and a simulation-to-field gap remains despite temporally correlated radio errors and per-run hardware offsets. The real-data sources contain small UAV counts and partial modalities: MILUV has three indoor robots with UWB ranging quality, forestry supplies motion without radio labels, AERPAW is UAV-to-infrastructure cellular, and WiNES supplies peer-link radio without a matching network controller.

The reported latency is a host-process loop, not sensor-to-autopilot or end-to-end vehicle latency, and no field deployment or embedded-hardware benchmark was performed. The radio model abstracts shadowing, fading, and correlation; the traffic model uses a simplified single collision domain, routing, queues, retransmission outcomes, and deadlines. The stateful relay remains a simulated intervention with fixed speed, acceleration, reaction, duration, and link-budget constraints. Consequently, no result establishes outdoor IP packet delivery, energy consumption, regulatory feasibility, multi-channel interference, safe autonomous actuation, or deployment readiness.

9. Reproducibility

The anonymous package contains configurations, source code, per-seed outputs, protocols, derived real-data traces, hashes, generated tables, and figures. The MILUV downloader records the official dataset DOI, archive identifier, archive MD5, member CRC values, and per-file SHA-256 values. A readiness

audit checks cache version, seed coverage, PyTorch backend provenance, event columns, validation/test separation, horizon and factorial grids, packet loads, latency stages, support-aware 60 GHz scope, MILUV transfer, PDF page counts, anonymization, ZIP CRC, and package freshness.

The included interactive evidence replay binds only to generated CSV/JSON outputs and supports operating-policy comparison. It is a reproducibility aid rather than an independent experiment or hardware-connected system.

10. Conclusion

This study establishes a reproducible benchmark and cross-domain stress test for fragmentation-event forecasting in unmanned aerial vehicle networks. Under the locked primary test, Source-Gated Kinetic-TopoGuard does not show supported event- F_1 superiority over Current-state ExtraTrees and produces more false alerts. The equal-learner factorial, residual audit, measured-topology transfer, and closed-loop packet study independently show that richer features, better calibration, or higher warning scores do not by themselves establish engineering benefit. The supported contribution is therefore the leakage-controlled evaluation framework, complete baseline accounting, explicit claim boundaries, and reproducible negative evidence. Outdoor synchronized radio, packet, onboard-timing, and actuation validation remains future work rather than a deployment result.

Data and code availability

Code and derived artefacts are available at <https://github.com/ErcanErkalkan/FANET-TopoGNN>; the cited Zenodo record identifies the archived v0.1.0-q1-compact release, not the current working-tree version [41]. Third-party datasets retain their original licences and DOI records.

CRedit authorship contribution statement

Ercan Erkalkan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have influenced the work reported in this article.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the author used OpenAI GPT-based tooling for language refinement, software-development assistance, and local reproducibility cross-checks. The author reviewed and verified all resulting manuscript and code changes, executed the reported analyses, and takes responsibility for the submitted work.

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